



Scalable Distributed Data Anonymization

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What will be discussed today

- Refresh K-Anonymity + L-Diversity
- The Mondrian algorithm (theory + Spark impl)
- Distributed Job Scheduling

What is K-Anonymity

- Problem statement:
Given person-specific field-structured data, produce a release of the data with scientific guarantees that the individuals who are the subjects of the data cannot be re-identified while the data remain practically useful

What is K-Anonymity

- Property to enforce:
the information for each person contained in the release cannot be distinguished from at least $k-1$ individuals whose information also appear in the release

What is K-Anonymity

Is this dataset k-anonymous for $k = 2$?

SSN	Name	Race	Date of birth	Sex	ZIP	Marital status	Disease
		asian	64/04/12	F	94142	divorced	hypertension
		asian	64/09/13	F	94141	divorced	obesity
		asian	64/04/15	F	94139	married	chest pain
		asian	63/03/13	M	94139	married	obesity
		asian	63/03/18	M	94139	married	short breath
		black	64/09/27	F	94138	single	short breath
		black	64/09/27	F	94139	single	obesity
		white	64/09/27	F	94139	single	chest pain
		white	64/09/27	F	94141	widow	short breath

What is K-Anonymity

The previous question is not meaningful, since it is not specified what parts of the table has to be 2-anonymous

SSN	Name	Race	Date of birth	Sex	ZIP	Marital status	Disease
		asian	64/04/12	F	94142	divorced	hypertension
		asian	64/09/13	F	94141	divorced	obesity
		asian	64/04/15	F	94139	married	chest pain
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		black	64/09/27	F	94139	single	obesity
		white	64/09/27	F	94139	single	chest pain
		white	64/09/27	F	94141	widow	short breath

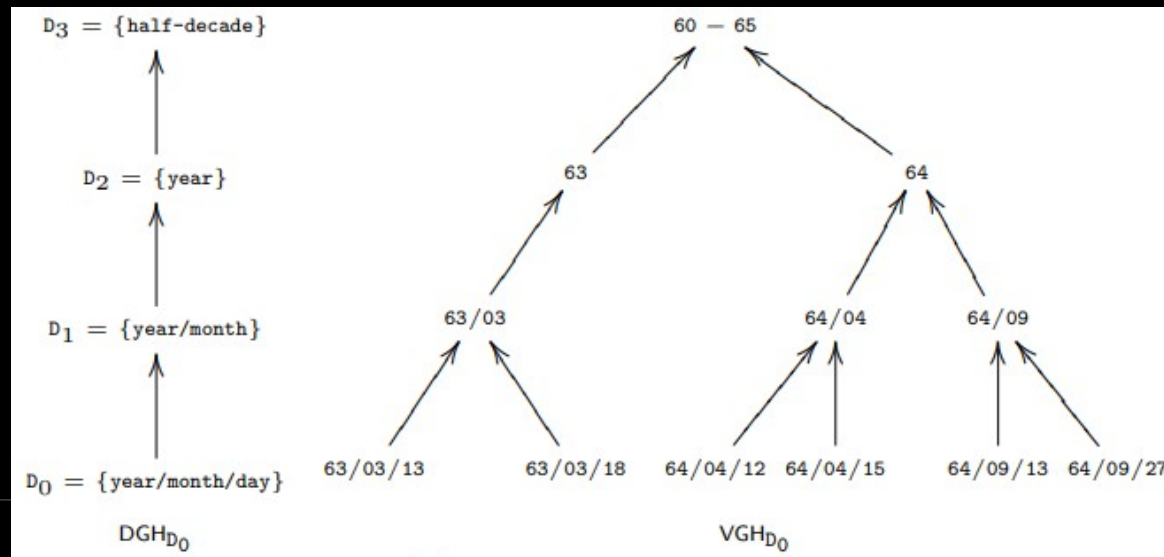
The attributes that can be used by an attacker to perform a linkage attack are called **quasi-identifiers**

Why such a model is necessary?

- Popular concerns:
 - Netflix published data about movie rankings for 500,000 customers in 2007, and researchers showed they could de-anonymize the data using a few additional inputs from IMDb
 - Using 1990 U.S. census data, Stanford researchers showed that they could uniquely identify **87 percent** of the U.S. population using only their Zip code, gender, and date of birth
 - AOL published search data for 650,000 users in 2007, thinking it was enough to anonymize their name using a unique ID. Unfortunately, most users often query their own name. As a result, their CTO resigned and an entire research team was fired after the public outcry

How to achieve K-Anonymity

- Suppression: remove the tuple from the dataset
- Generalization: substituting the values of a given attribute with more general values



Introducing L-Diversity

- Some attributes can be valuable as much as identifiers
- K-Anonymity does not enforce any constraint on the cardinality of the set of values in a group of K individuals

Introducing L-Diversity

	Non-Sensitive			Sensitive
	Zip Code	Age	Nationality	Condition
1	130**	< 30	*	Heart Disease
2	130**	< 30	*	Heart Disease
3	130**	< 30	*	Viral Infection
4	130**	< 30	*	Viral Infection
5	1485*	≥ 40	*	Cancer
6	1485*	≥ 40	*	Heart Disease
7	1485*	≥ 40	*	Viral Infection
8	1485*	≥ 40	*	Viral Infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

The Mondrian algorithm

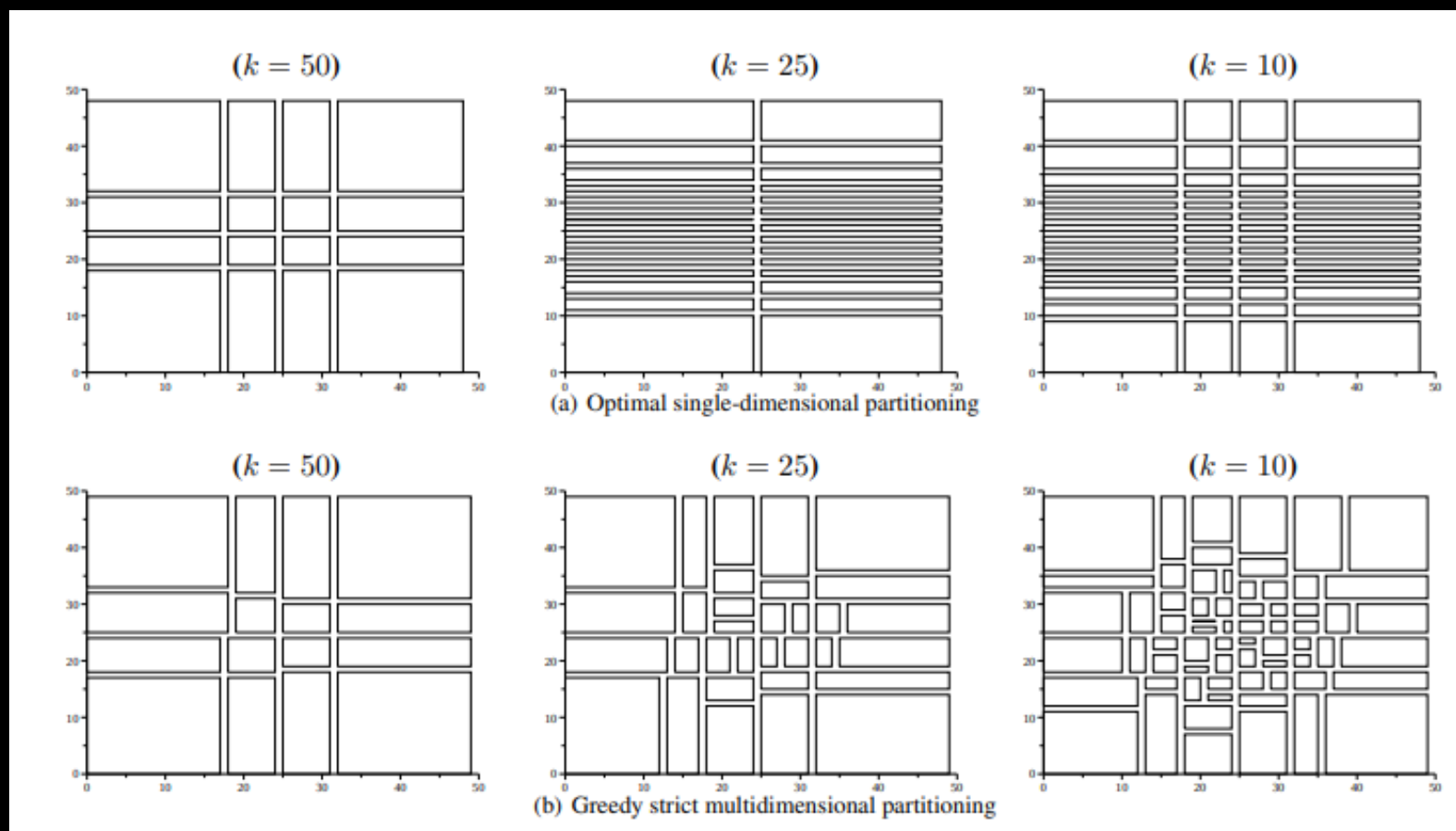
- Greedy approach to enforce K-Anonymity + L-Diversity
- More efficient than the optimal single attribute approach ($n\log(n)$ vs $\exp(n)$)
- Reported to produce better results in the multidimensional scenario

The Mondrian algorithm

```
Anonymize(partition)  
  if (no allowable multidimensional cut for partition)  
    return  $\phi : \textit{partition} \rightarrow \textit{summary}$   
  else  
     $\textit{dim} \leftarrow \text{choose\_dimension}()$   
     $\textit{fs} \leftarrow \text{frequency\_set}(\textit{partition}, \textit{dim})$   
     $\textit{splitVal} \leftarrow \text{find\_median}(\textit{fs})$   
     $\textit{lhs} \leftarrow \{t \in \textit{partition} : t.\textit{dim} \leq \textit{splitVal}\}$   
     $\textit{rhs} \leftarrow \{t \in \textit{partition} : t.\textit{dim} > \textit{splitVal}\}$   
    return Anonymize(rhs)  $\cup$  Anonymize(lhs)
```

- Strict version: lhs and rhs have no values in common

Lesson learned: visualization is important



Mondrian components

- `choose_dimension()` → normalized span
- `frequency_set()` → scan
- `find_median()` → sort and filter
- `hidden` => check partition validity → scan

Reality check – Problems of Mondrian

- `choose_dimension()` $\rightarrow O(M * N)$
 - `frequency_set()` $\rightarrow O(N)$
 - `find_median()` $\rightarrow O(F \log(F))$
 - `hidden` $\Rightarrow$ check partition validity $\rightarrow O(1)$ for K-Anonymity, $O(K * N)$ for L-Diversity
-
- N = tuples in the partition • F = # unique values
 - M = # of quasi-identifiers • K = # of sensitive attributes

Reality check – Problems of Mondrian

- `choose_dimension()` → A lot of quasi-identifiers?
- `frequency_set()` → Large datasets?
- `find_median()` → $F \rightarrow N$?
- What about attributes without order?

Spark to the rescue

DEMO TIME!

Repository link: <https://github.com/mosaicrown/mondrian>

Papers links:

<https://spdp.di.unimi.it/papers/percom2021.pdf>

<https://spdp.di.unimi.it/papers/percom2021-artifact.pdf>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9894678>

Job Scheduling – Final push for today

- Problem of normal job scheduling?
- Problem of job scheduling over a cluster?
- Any idea for the architecture to use?

Credits

These slides were originally written by Gianluca Oldani for the course of Advanced Data Management held for the academic year 2022 / 2023